

Project Report

TITLE: Cosmetic Insights: Navigating Cosmetics Trends and Consumer Insights with Tableau

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1. INTRODUCTION

1.1 Project Overview

The cosmetics industry is one of the fastest-growing and highly competitive sectors, driven by changing consumer preferences, diverse skin types, brand innovation, and evolving pricing strategies. With increasing demand for personalised beauty products, companies rely on data-driven insights to understand market trends, consumer behaviour, and product performance.

This project, “Cosmetic Insights: Navigating Cosmetics Trends and Consumer Behaviour with Tableau,” focuses on analysing a cosmetics dataset to uncover meaningful patterns and actionable insights. The dataset includes key attributes such as brand, product category, price, rank, and skin type suitability (Dry, Oily, Combination, Normal, and Sensitive).

This project is designed to help businesses and analysts understand cosmetics market trends and consumer preferences. By analysing brand performance, pricing patterns, product distribution, and skin-type coverage, the dashboard enables stakeholders to identify high-performing products, target specific customer segments, and optimise marketing and product strategies.

1.2 Purpose

The purpose of this Tableau project is to convert raw cosmetics data into an interactive, visual decision-making tool that helps beauty brands quickly adapt to changing market trends and consumer needs. Instead of looking at static spreadsheets, businesses can use these dashboards to spot risks instantly and find new product opportunities.

Key Objectives & Business Purpose

- **Speed Up Decisions:** Moves companies from reactive guessing to proactive action. For example, instead of waiting for quarterly sales reports, teams can see a sudden drop in product interest within days and adjust marketing immediately.
- **Protect Brand Reputation:** Acts as an early warning system for product issues. Brands can isolate bad batches or poorly reviewed formulations instantly, allowing them to fix quality control issues before customer trust drops.
- **Identify Market Gaps:** Pinpoints exactly which skin types or product categories are underserved. R&D teams can see clear visual "white spaces" in the market to design new, highly targeted products.
- **Optimise Pricing Strategies:** Reveals the sweet spot between product price and consumer satisfaction (rank). This helps brands see if they are underpricing a high-value product or overpricing an item that consumers dislike.

2.Ideation Phase

2.1 Problem Statement

Create a problem statement to understand your customer's point of view. The Customer Problem Statement template helps you focus on what matters to create experiences people will love.

A well-articulated customer problem statement allows you and your team to find the ideal solution for the challenges your customers face. Throughout the process, you'll also be able to empathise with your customers, which helps you better understand how they perceive your product or service.

I am	Describe customer with 3-4 key characteristics - who are they?	Describe the customer and their attributes here
I'm trying to	List their outcome or "job" the care about - what are they trying to achieve?	List the thing they are trying to achieve here
but	Describe what problems or barriers stand in the way - what bothers them most?	Describe the problems or barriers that get in the way here
because	Enter the "root cause" of why the problem or barrier exists - what needs to be solved?	Describe the reason the problems or barriers exist
which makes me feel	Describe the emotions from the customer's point of view - how does it impact them emotionally?	Describe the emotions the result from experiencing the problems or barriers

Example:

I am a traveler	I'm trying to book flights on my phone	But it takes a long time	Because The website is not responsive and doesn't have a mobile version	Which makes me feel Frustrated
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Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	An online shopper	Return a pair of shoes that do not fit	I cannot find a return shipping label anywhere on the website	The digital return portal requires contacting customer service first	Annoyed and trapped
PS-2	A patient	Log my daily	The	The	Overwhelmed and

	tracking a chronic condition	medication dose	logging process takes too many taps and screen transitions	interface requires manual data entry instead of quick presets	discouraged
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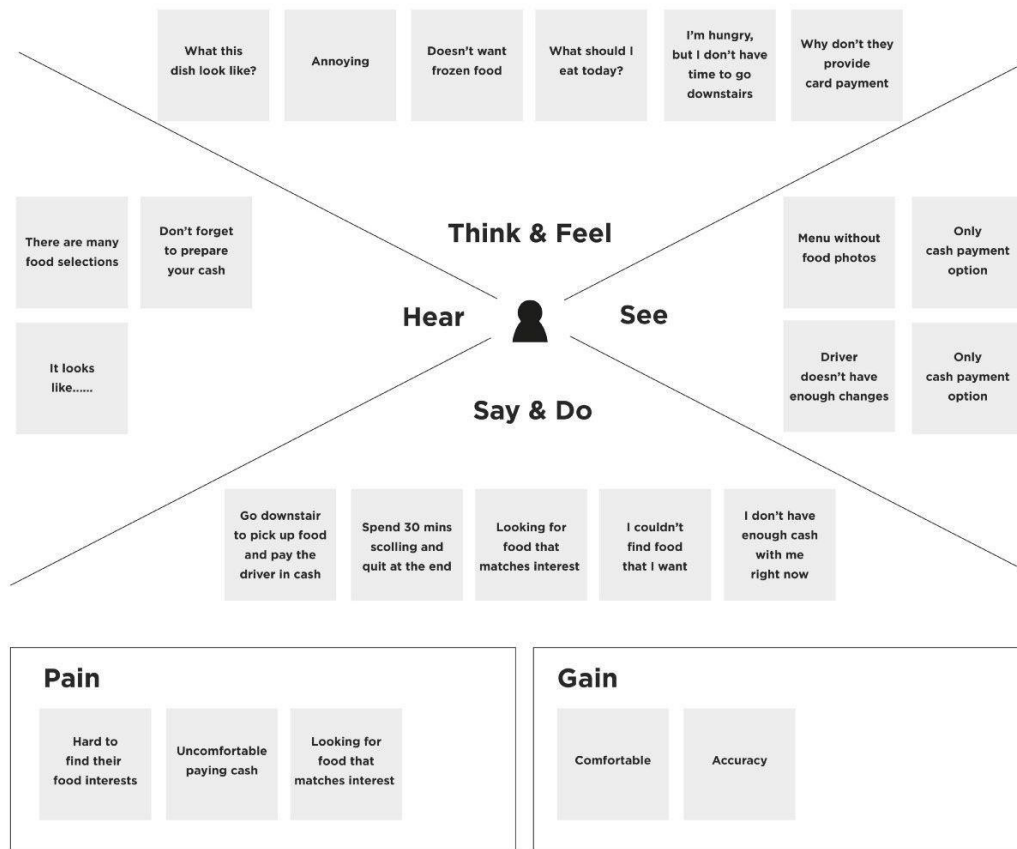
2.2 Empathy Map:

An empathy map is a simple, easy-to-digest visual that captures knowledge about a user's behaviours and attitudes.

It is a useful tool to help teams better understand their users.

Creating an effective solution requires understanding the true problem and the person who is experiencing it. The exercise of creating the map helps participants consider things from the user's perspective along with his or her goals and challenges.

Example:



2.3 Brainstorming

Brainstorming provides a free and open environment that encourages everyone within a team to participate in the creative thinking process that leads to problem solving. Prioritising volume over value, out-of-the-box ideas are welcome and built upon, and all participants are encouraged to collaborate, helping each other develop a rich amount of creative solutions.

Use this template in your own brainstorming sessions so your team can unleash their imagination and start shaping concepts even if you're not sitting in the same room.

Step-1: Team Gathering, Collaboration and Select the Problem Statement



Brainstorm & idea prioritization

Use this template in your own brainstorming sessions so your team can unleash their imagination and start shaping concepts even if you're not sitting in the same room.

 10 minutes to prepare
 1 hour to collaborate
 2-8 people recommended

 **Before you collaborate**

A little bit of preparation goes a long way with this session. Here's what you need to do to get going.

 10 minutes

A Team gathering
Define who should participate in the session and send an invite. Share relevant information or pre-work ahead.

B Set the goal
Think about the problem you'll be focusing on solving in the brainstorming session.

C Learn how to use the facilitation tools
Use the Facilitation Superpowers to run a happy and productive session.

[Open article](#) →

1 Define your problem statement

What problem are you trying to solve? Frame your problem as a How Might We statement. This will be the focus of your brainstorm.

 5 minutes

PROBLEM

How might we [your problem statement]?

Key rules of brainstorming

To run a smooth and productive session

 Stay in topic.	 Encourage wild ideas.
 Defer judgment.	 Listen to others.
 Go for volume.	 If possible, be visual.

Step-2: Brainstorm, Idea Listing and Grouping

2

Brainstorm

Write down any ideas that come to mind that address your problem statement.

🕒 10 minutes

TIP

You can select a sticky note and hit the pencil (switch to sketch) icon to start drawing!

Amar

Yuktesh

Person 3

Person 4

Person 5

Person 6

Person 7

Person 8

3

Group ideas

Take turns sharing your ideas while clustering similar or related notes as you go. In the last 10 minutes, give each cluster a sentence-like label. If a cluster is bigger than six sticky notes, try and see if you can break it up into smaller sub-groups.

🕒 20 minutes

TIP

Add customizable tags to sticky notes to make it easier to find, browse, organize, and categorize important ideas as themes within your mural.

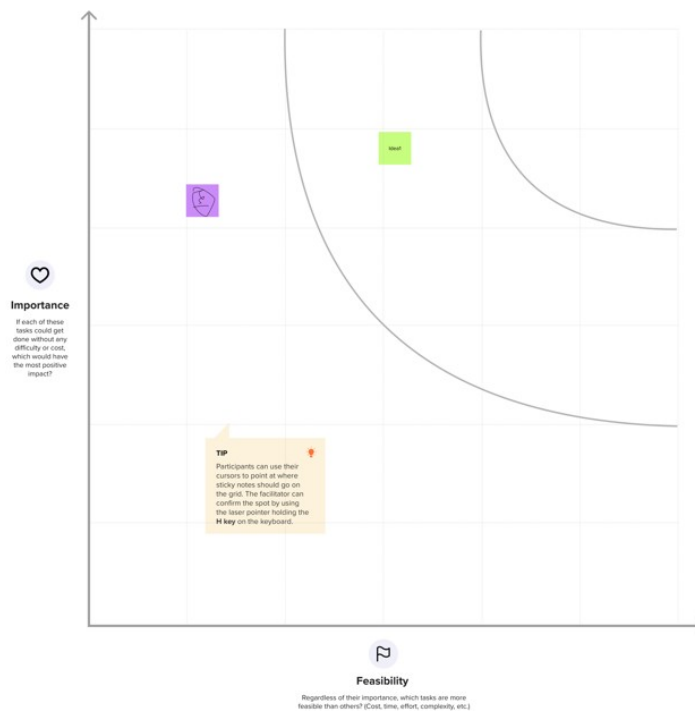
Step-3: Idea Prioritization

4

Prioritize

Your team should all be on the same page about what's important moving forward. Place your ideas on this grid to determine which ideas are important and which are feasible.

🕒 20 minutes



3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

CUSTOMER JOURNEY MAP				
AWARENESS	CONSIDERATION	DECISION	ACTION	POST-ACTION
Customer Goal	Explore tools to visualize and track student nutrition patterns	Choose a data visualization platform to monitor and improve dietary trends	Implement visual dashboards and monitor student data	Evaluate impact, adjust strategies, and plan next interventions
Touchpoints	Research on tools like Tableau, workshops, health committee meetings	Tableau trials, case studies, peer recommendations	Tableau dashboards, KPI alerts, student reports	Follow-up dashboards, wellness program feedback
Experience	Curious but overwhelmed by data complexity	Confident in Tableau's interactive dashboard capabilities	Empowered to make decisions based on trends and alerts	Reflects on dashboard insights and makes recommendations for next semester
Pain Points	No integrated system for diet tracking	Budget limitations, training needed for Tableau	Time consuming data cleaning, interpreting visualizations	Difficulty measuring long-term change
Opportunities	Educate stakeholders about dietary data value	Show past success stories and predicted impact	Automate data updates, scheduled alerts	Expand dashboards with advanced analytics and student-level personalization

3.2 Solution Requirement

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	Registration through Form Registration through Gmail Registration through LinkedIn
FR-2	User Confirmation	Confirmation via Email Confirmation via OTP
FR-3	Data Visualisation	Generate Text table Generate Horizontal bar Chart Generate side-by-side bar Generate Pie Chart Generate Histogram Chart Generate packed bubble sort
FR-4	Data Filtering and Sorting	Filter by Data Range, Measure Names Sort by Calculations, Fit by Entire view

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	The system should have a user-friendly and intuitive interface.
NFR-2	Security	Secure login and user data encryption should be implemented.

NFR-3	Reliability	The system should perform consistently under expected loads without failure.
NFR-4	Performance	Visualisations should load within 3 seconds on average.
NFR-5	Availability	The system should be accessible 24/7 with minimum downtime.
NFR-6	Scalability	Should be scalable to handle large datasets or more branches in the future.

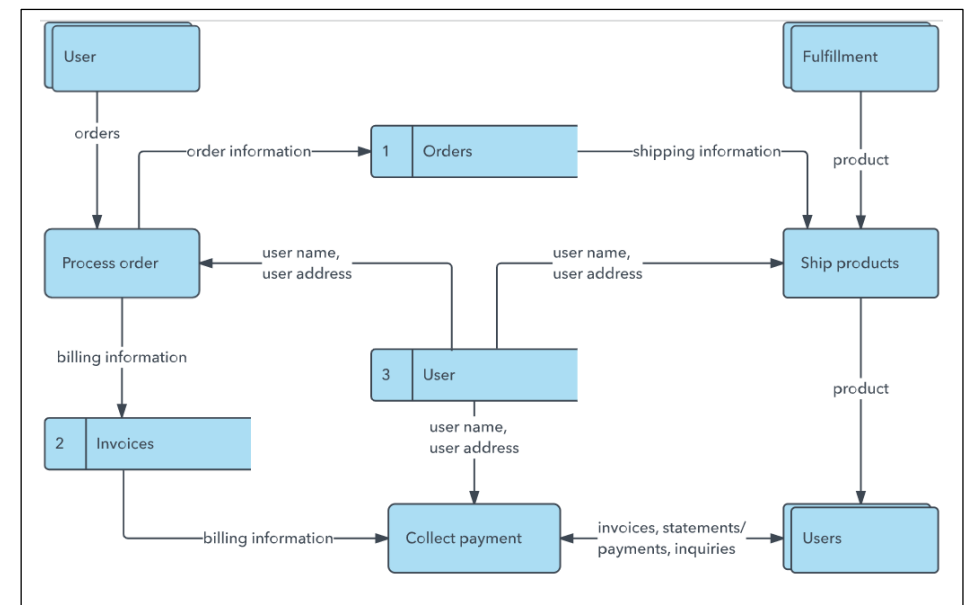
3.3 Data Flow Diagram

Data Flow Diagrams:

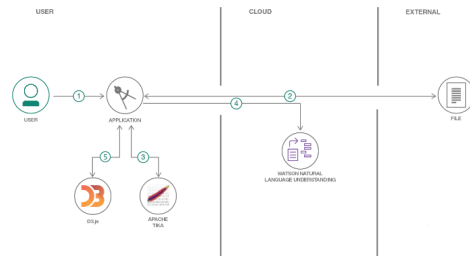
A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

Example: [\(Simplified\)](#)

Example: DFD Level 0 (Industry Standard)



Flow



1. User configures credentials for the Watson Natural Language Understanding service and starts the app.
2. User selects data file to process and load.
3. Apache Tika extracts text from the data file.
4. Extracted text is passed to Watson NLU for enrichment.
5. Enriched data is visualized in the UI using the D3.js library.

User Stories

Use the below template to list all the user stories for the product.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Customer (Mobile user)	Registration	USN-1	As a user, I can register for the application by entering my email, password, and confirming my password.	I can access my account / dashboard	High	Sprint-1
Customer (Mobile user)	Registration	USN-2	As a user, I will receive confirmation email once I have registered for the application	I can receive confirmation email & click confirm	High	Sprint-1
Customer (Mobile user)	Registration	USN-3	As a user, I can register for the application through Facebook	I can register & access the dashboard with Facebook Login	Low	Sprint-2

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Customer (Web user)	Registration	USN-4	As a user, I can register for the application through Gmail	I can register & access the dashboard with Gmail Login	Medium	Sprint-1
Customer (Web user)	Login	USN-5	As a user, I can log into the application by entering email & password	I can successfully log in and view my account dashboard	High	Sprint-1
Customer (Web user)	Dashboard	USN-6	As a web user, I can view my profile data, recent activities, and account metrics on the main dashboard	Dashboard displays accurate profile info and recent activity data within 2 seconds	High	Sprint 1
Customer Care Executive	Ticket Management	USN-7	As a support agent, I can view and update customer support tickets	I can search tickets by ID, update statuses, and add internal notes	High	Sprint 2
Administrator	User Management	USN-8	As an admin, I can view, suspend, or activate user accounts	Admin portal displays full user lists with functional toggles to activate/deactivate accounts	Medium	Sprint 2

3.4 Technology Stack

Data Layer

Component	Description
Dataset	Cosmetics.csv – structured Csv dataset capturing cosmetics trends and consumer info
Storage Format	Flat file (CSV) loaded locally for analysis
Tools Used	Excel (for cleaning & preprocessing if needed)

Data Processing Layer

Tool/Technology	Purpose
Excel	Initial cleaning or quick field review before importing to Tableau

Visualization & Analytics Layer

Tool/Technology	Purpose
Tableau	Main tool for interactive data visualization and dashboard creation
Tableau Features	KPI cards, Bar Charts, Pie Charts, Graphs, Highlight Tables, Forecasts
Calculated Fields	Used for: Consumption, Cosmetics selection

User Interaction Layer

Feature	Role
Interactive Dashboards	Users can filter data by Labels and Brand
Parameter Controls	Customize target values
Alerts & KPIs	Instant insight into total counts and trends
User Roles	Admin, Consumers ,Customers – each interacts with filtered views

Security & Sharing

Feature	Notes
Tableau Public	Public dashboards (for non-sensitive data)
Tableau Server	Optional upgrade for secured, role-based access
Export Options	PDF reports, public link sharing, dashboard embedding

Justification for Technology Choices

- **Tableau** is chosen for its ease of use, powerful visualization capabilities, and ability to integrate dynamic, calculated insights without extensive coding.
- **CSV Format** keeps the project simple and lightweight for academic or prototype environments.
- **Excel/Python** enables initial analysis and advanced preprocessing before visual design begins.

4. PROJECT DESIGN

4.1 Problem Solution Fit

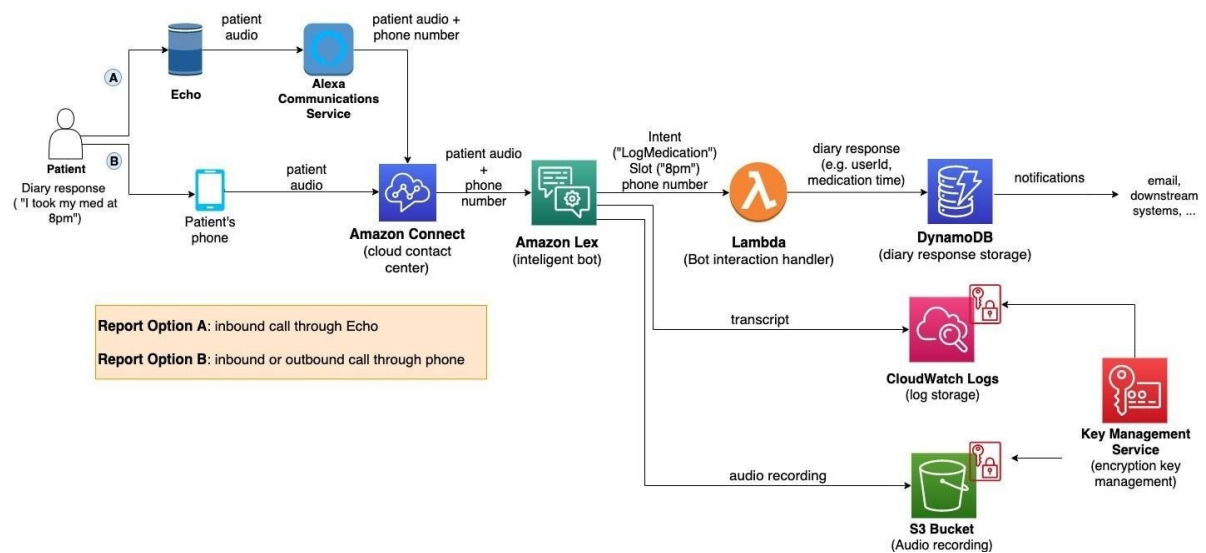
1. CUSTOMER SEGMENT(S) CS College students and young working professionals aged 18-30 who are health-conscious but have poor diet routines due to lack of time, awareness, or planning tools. Define CS, fit into	4. CUSTOMER CC Low attention span or time Limited money for professional nutritionists No access to real-time food feedback Device limitations (some don't use laptops, only mobile) Overwhelmed by technical terms	6. AVAILABLE SOLUTIONS AS Mobile health apps (MyFitnessPal, HealthilyMe) YouTube diets / Instagram influencers. Envision AS
2. JOBS TO BE DONE / PROBLEMS JD Want to eat healthy but don't know how to start. Don't have time to plan meals or analyze food choices. Struggle to track nutrients or calories effectively. Feel overwhelmed by too much inconsistent advice online. Want to avoid lifestyle diseases but have no support system. Focus on JD, fit into BC, understand	3. PROBLEM ROOT CAUSE PRC Most customers lack structured, visual, and reliable food habit tracking tools tailored to their age/lifestyle. School and college curriculums often don't include practical nutrition education, leading to ignorance. They want to change, but the complexity of diet science makes it hard to start or stay consistent.	7. BEHAVIOUR BD Watch diet tips online Download free meal planners Try calorie counting apps Browse Google for quick-fix diets Sometimes buy health food impulsively Focus on BD, fit into BC, understand
3. TRIGGERS TR Weight gain or health issues (e.g., acne, fatigue). Social media influence (seeing fitness transformations). Doctor advice to follow a better diet. 4. EMOTIONS: BEFORE / AFTER EM Confused, guilty, overwhelmed, procrastinating Empowered, in control, health-aware, optimistic Identify strong TR & EM	10. YOUR SOLUTION SL Analyzes historical food choice data Provides personalized food insights by gender, age, lifestyle Visualizes dietary patterns and gaps Suggests realistic meal changes	8. CHANNELS of BEHAVIOUR CH 1.1 ONLINE Follow diet influencers on Instagram or YouTube 1.2 OFFLINE Talk to gym instructors or peers Try following roommates' diets Expand online & offline CH of BD

4.2 Proposed Solution

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Clearly describe the problem you aim to solve. Include relevant statistics or pain points to highlight the severity or urgency of the issue.
2.	Idea / Solution description	Outline your proposed solution. Focus on how it addresses the problem effectively and efficiently.
3.	Novelty / Uniqueness	What makes your solution different from existing ones? Highlight any innovative features, technologies, or processes.
4.	Social Impact / Customer Satisfaction	Explain how your solution improves lives, benefits communities, or enhances customer satisfaction.
5.	Business Model (Revenue Model)	How will the solution generate revenue? Include pricing strategies, partnerships, target customer segments, etc.
6.	Scalability of the Solution	Discuss how your solution can grow geographically or serve more users. Mention potential challenges and how they will be handled.

4.3 Solution Architecture

Example-Solution Architecture Diagram:



5. PROJECT PLANNING & SCHEDULING

5.1 Project planning

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Registration	USN-1	As a user, I can register for the application by entering my email, password, and confirming my password.	2	High	5
Sprint-1	Confirmation	USN-2	As a user, I will receive confirmation email once I have registered for the application	1	High	5
Sprint-2	Registration	USN-3	As a user, I can register for the application through Facebook	2	Low	5
Sprint-1	Registration	USN-4	As a user, I can register for the application through Gmail	2	Medium	5
Sprint-1	Login	USN-5	As a user, I can log into the application by entering email & password	1	High	5

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	20	6 Days	01 June 2026	08 June 2026	20	08 June 2026
Sprint-2	20	6 Days	09 June 2026	15 June 2026	20	15 June 2026
Sprint-3	20	6 Days	16 June 2026	22 June 2026	20	22 June 2026
Sprint-4	20	6 Days	23 June 2026	30 June 2026	20	30 June 2026

Velocity:

Imagine we have a 10-day sprint duration, and the velocity of the team is 20 (points per sprint). Let's calculate the team's average velocity (AV) per iteration unit (story points per day)

$$AV = \frac{\text{sprint duration}}{\text{velocity}} = \frac{20}{10} = 2$$

5.2 Planning Logic

A Sprint fixed period or duration in which a team works to complete a set of tasks

An **Epic** is a **big task or project** that is too large to complete in one sprint. It is broken down into **smaller tasks (stories)** that can be completed over multiple sprints.

A **Story** is a small task . It is part of an **Epic**.

A **Story Point** is a number that represents how much effort a story takes to complete.
(usually in form of Fibonacci series)

- 1- Very Easy task
- 2- Normal task
- 3- Moderate task
- 5- Difficult task

Storypoint -5 (1,2,3,5)

Sprint 1

Data Collection (Epic 1)

Gathering Data (USN1) 2

Loading Data(USN2) 1

Data Preparation (Epic 2)

Handling Missing Values(USN3) 3

Creating Fields(USN4)3

Handling Inconsistency in Data(USN5)3

Total Story Point in Sprint 1= 2+1+3+3+3=12

Sprint 2

Data Visualization (Epic 3)

Bar Chart(USN6) 2

Pie Chart(USN7) 2

Line Chart(USN8) 2

Map(USN9) 4

Dashboard (Epic 4)

Developing Dashboard(USN10) 5

Story (Epic 5)

Developing Story(USN11) 5

Total story Point in Sprint 2= 2+2+2+4+5+5=20

Total Story Points

Sprint 1 = 12

Sprint 2 = 20

Velocity= Total Story Points Completed/ Number of Sprints

Total story Points= 12+20=32

No of Sprints= 2

Velocity =32/2

16(Story Points per Sprint)

Your team's velocity is 16 Story Points per Sprint.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

S.No.	Parameter	Screenshot / Values
1.	Data Rendered	1472 records
2.	Data Preprocessing	Removed null values
3.	Utilization of Filters	Date Range Filter Region Filter Product Category Filter
4.	Calculation fields Used	Usually numeric, date/time, or text depending on the formula and context.
5.	Dashboard design	No of Visualizations / Graphs -1
6	Story Design	No of Visualizations / Graphs -1

Story 1

Category Level	Category Wise total revenue	Top 10 Brands that are suitable	Top 10 products by product count	Brand with there averaging pricing	Category wise distribution of	Top 10 Brands with cosmetics	Category Level	KPI's used in this Data	Cosmetics Insights
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Legend:

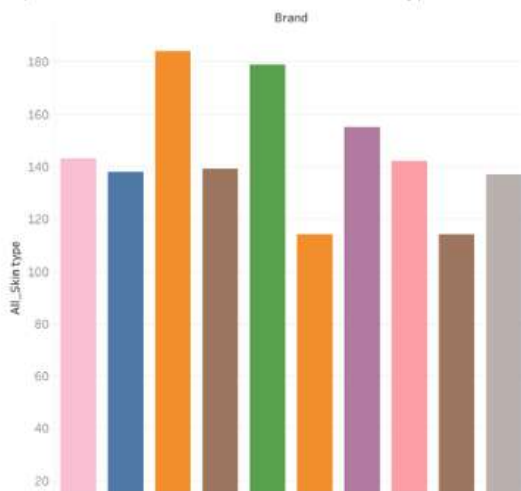
- Cleanser
- Eye cream
- Face Mask
- Moisturizer
- Sun product
- Treatment



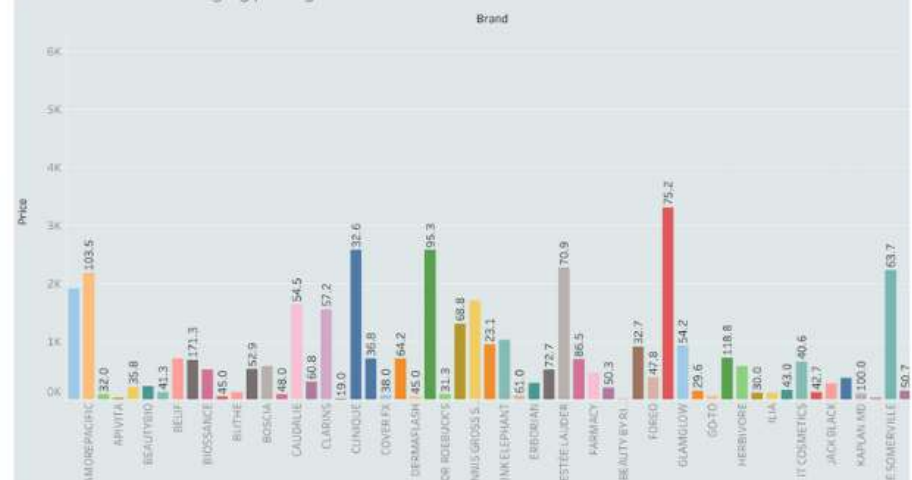
7. RESULTS

7.1 Output Screenshots

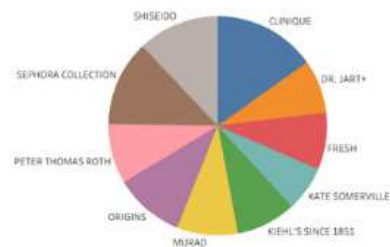
Top 10 Brands that are suitable for all skin types



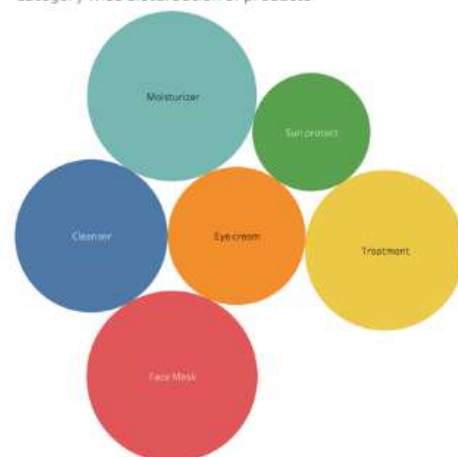
Brand with there averaging pricing

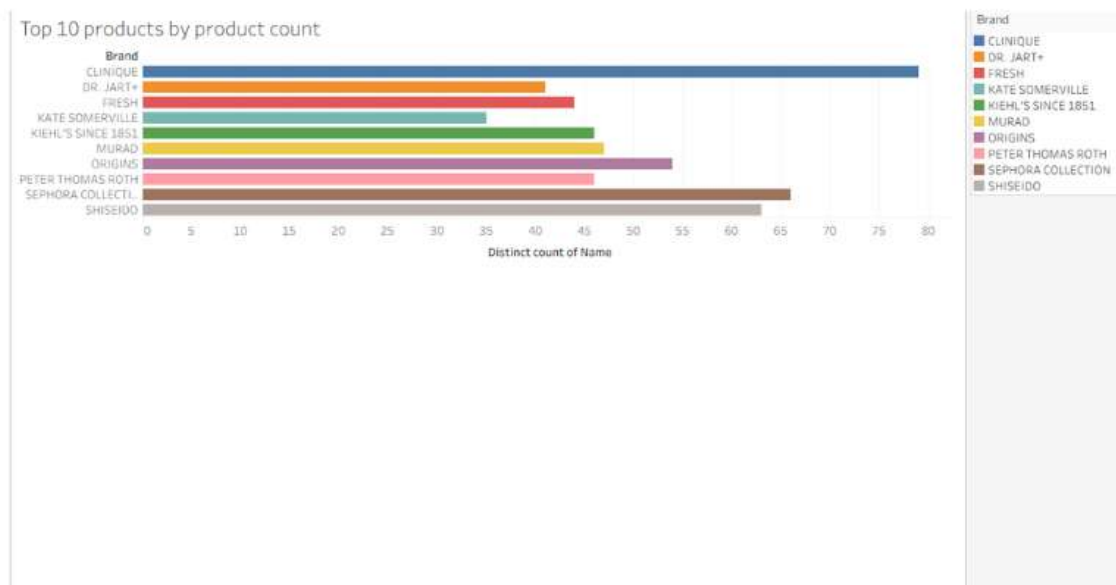
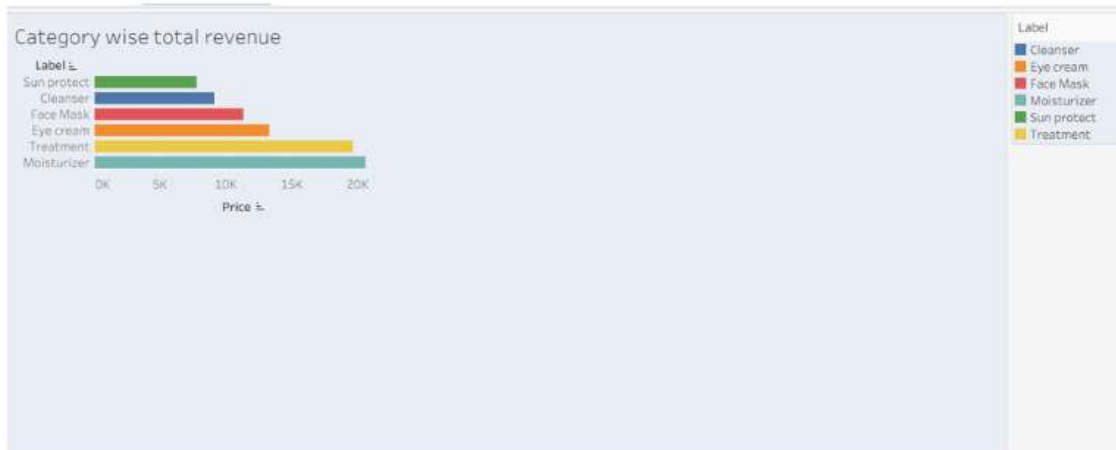


Top 10 Brands with cosmetics count



Category wise distubution of products





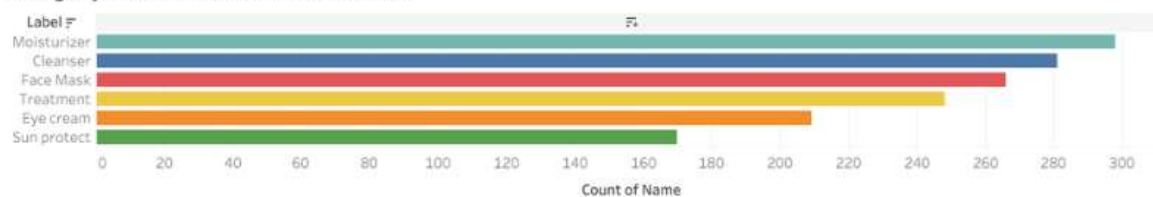
Category-Level Product Distribution Analysis



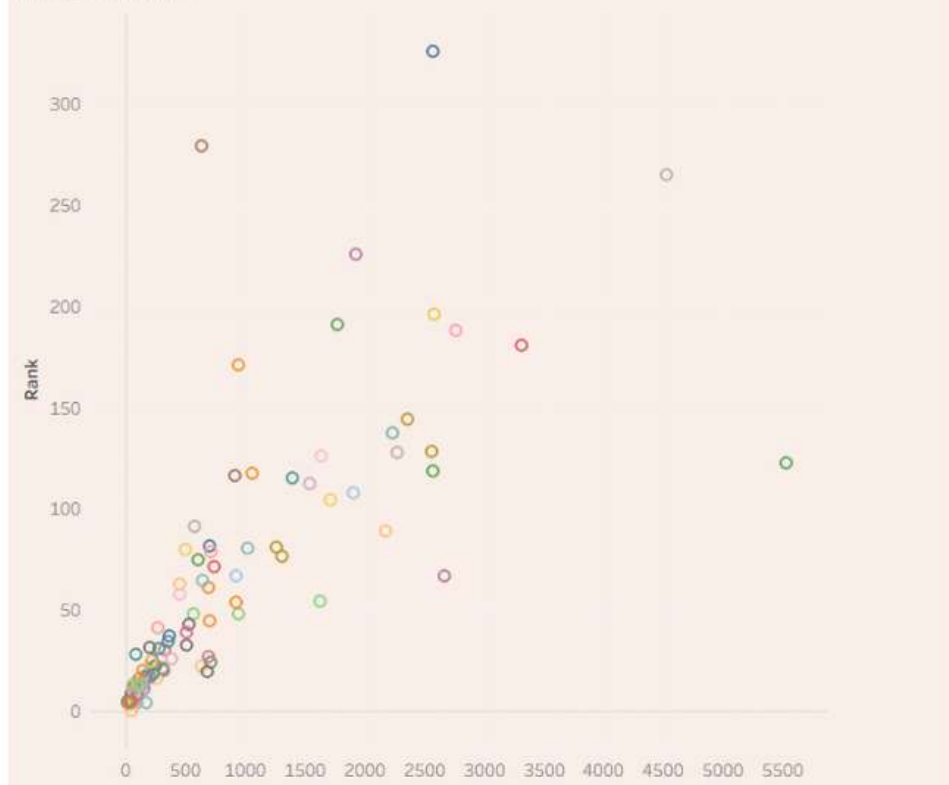
Label

- Cleanser
- Eye cream
- Face Mask
- Moisturizer
- Sun protect
- Treatment

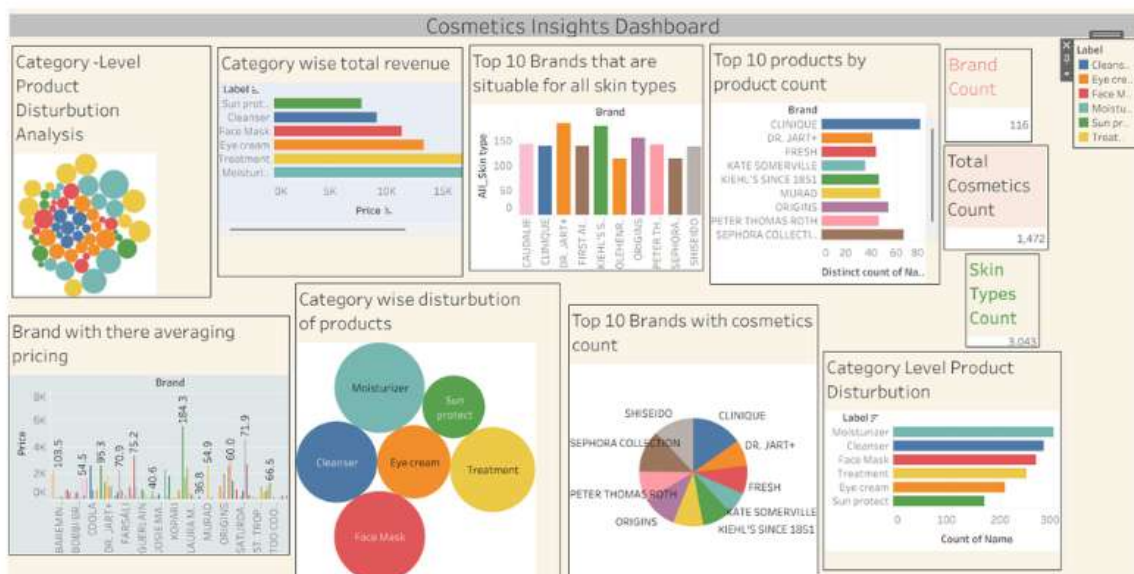
Category Level Product Distribution



Price vs Rank



Dashboard



Story

Story 1



ADVANTAGES AND DISADVANTAGES OF COSMETICS INSIGHTS

Advantages

- **Rapid Trend Identification:** Integrates social media buzz, web search volume, and consumer reviews. This allows brands to quickly pivot from traditional makeup lines to rising demands like clinical skincare, neuro-cosmetics, or metabolic beauty.
- **Hyper-Personalisation:** Segments consumers based on purchase history, skin types, and browsing habits. This empowers brands to run precision marketing campaigns and send tailored product recommendations.
- **Real-Time Inventory Management:** Replaces retroactive reporting with live data. It maps exactly where online traffic drops off, tracks which localised shade assortments are selling best, and flags inventory bottlenecks.

Disadvantages

- **High Cost & Setup Complexity:** Tableau is one of the most expensive business intelligence platforms. For smaller indie beauty brands, the total cost of ownership—including data engineering and licensing—can be prohibitive.
- **Data Preparation Limitations:** Tableau requires clean, organised datasets. It lacks heavy data cleansing and preparation tools, forcing teams to rely on external software (like Alteryx or SQL) before bringing data into Tableau.
- **Static Parameters:** Unlike more flexible coding alternatives (e.g., Python), Tableau's dashboard parameters are somewhat rigid. This means parameters like seasonal shade ranges or localized price changes often need manual, single-value updates.
- **No Automated Reporting/Scheduling:** Native report-scheduling and automatic refreshing features are limited without deploying Tableau Cloud or Server, which adds more overhead.

CONCLUSION

Summary of Project Outcomes

This project successfully transformed raw beauty industry data into a dynamic, data-driven visual asset. By leveraging Tableau, the project bridged the gap between complex consumer behaviour and actionable retail strategy. It shifted the brand approach from reactive guessing to proactive, data-backed forecasting.

Core Project Conclusions

- **Trend Responsiveness:** Tableau's real-time filtering proved that tracking fast-moving viral beauty trends (such as clean beauty or specific active ingredients) allows brands to reduce product launch times and capitalize on market demand instantly.
- **Consumer-Centric Design:** Demographic and sentiment analysis maps revealed distinct geographic preferences. This allows for hyper-localized marketing campaigns and regional product distribution instead of a "one-size-fits-all" approach.
- **Optimized Product Mix:** Interactive dashboards successfully isolated low-performing product lines and high-margin hero items. This clear visibility empowers stakeholders to optimize inventory and phase out stagnant stock.
- **Enhanced Decision-Making:** Moving away from static spreadsheets to interactive visual analytics reduced data synthesis time for stakeholders. This shift allows executives to identify emerging market shifts in seconds rather than days.

Final Project Takeaway

Ultimately, "**Cosmetic Insights**" demonstrates that data visualization is no longer optional in the highly competitive beauty landscape. By utilizing Tableau to synthesize consumer sentiments, purchasing habits, and shifting trends, businesses can confidently mitigate risks, maximize marketing ROI, and build stronger, more personalized connections with their global consumer base.

FUTURE SCOPE

The future scope of cosmetic insights focuses on five major pillars:

1. Biological Innovation: The industry is moving from chemistry to biology. Brands are investing in biotechnology to engineer ingredients that provide specific, cellular-level results (e.g., collagen banking and barrier repair).

2. AI and Hyper-Personalization: Artificial Intelligence is heavily integrated from Research & Development to the digital shelf. AI is used to formulate custom skin and hair care profiles based on individual genetics, localized environmental factors, and historical usage trends.

3. Predictive Market Analytics: Top firms track retail transaction logs, social media trend velocity, and global search ranks to predict market shifts in real-time. Platforms like [NielsenIQ Beauty](#) utilize predictive analytics to measure ingredient-level product characteristics and competitive gaps.

4. Evidence-Based Claims: Savvy consumers require scientific proof for product effectiveness across makeup and skincare. Brands are leaning on dermatological partnerships and clinical trials to ensure radical honesty and transparency.

5. Wellness & Holistic Integration: Consumers desire routines that blend beauty with emotional well-being and longevity. This expands insights into adjacent categories like ingestible supplements, functional nutrition, and advanced at-home aesthetic tools (e.g., LED masks and radio frequency devices).

11. APPENDIX

Dataset Link:

<https://www.kaggle.com/datasets/kingabzpro/cosmetics-datasets>

GitHub Repository Link:

[**https://github.com/maheyb72-dev/Cosmetic-Insights-Navigating-Cosmetics-Trends-and-Consumer-Insights-with-Tableau**](https://github.com/maheyb72-dev/Cosmetic-Insights-Navigating-Cosmetics-Trends-and-Consumer-Insights-with-Tableau)

Project Demo Link:

[**https://drive.google.com/file/d/1H-Uf7SMMhMAIFFI03pepwbGZPlMPMKvx/view?usp=sharing**](https://drive.google.com/file/d/1H-Uf7SMMhMAIFFI03pepwbGZPlMPMKvx/view?usp=sharing)

CHANDAKA BINU MAHEIN